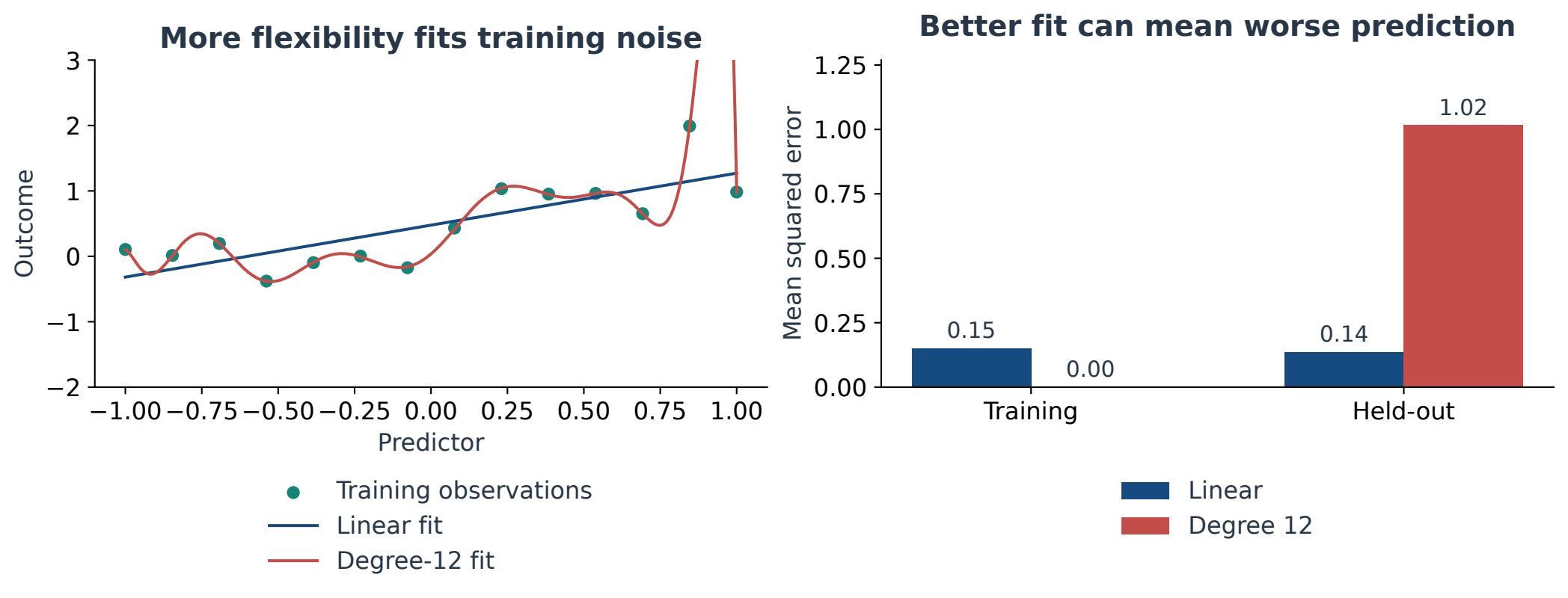
More flexibility fits training noise
Outcome
Predictor
3
2
1
0
−1
−2
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
Training observations
Linear fit
Degree-12 fit
Better fit can mean worse prediction
Mean squared error
1.25
1.00
0.75
0.50
0.25
0.00
0.15
0.00
0.14
1.02
Training
Held-out
Linear
Degree 12